day-06

Short Question Assignment:

Question 1: Jacobin Dimensionality

A single dense layer in a NN, takes an input X of size 512 and outputs an activation vector y of size 128.

What the exact Geometric dimension of Jacobin Matrix.

Solution:

The Jacobian Matrix will be (128×512)

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Question 2: The symmetric superpower

A) What mathmatical Th^m grantees that $\frac{\partial^2 L}{\partial w_1 \partial w_2} = \frac{\partial^2 L}{\partial w_2 \partial w_1}$

B) Bcz the Hessian is symmetric, what does the spectral Theorm (from day 3) guarantee about its eigenvalue & eigen vectors?

Solution:

A) Clairaut's Theorem (Schwarz Theorem)

B) The Spectral Theorem guarantees that the hessian will always have strictly real eigenvalues and perfectly orthogonal eigenvector.

Question 3: Classifying the critical point

You compute the hessian matrix at this exact point and find its eigenvalues to be $\lambda = [3.2, -1.1, 0.4, 2.8]$.

Are at local, min, max or saddle point?

Solution:

You are at a saddle point, The eigen value have mixed sign (positive, negative). This means the surface curves upwards in three independent

dimension and downwards in one.

Question 4: Second order optimization (Newton method)

If Newton method is mathmatically superior and requires for fewer steps to converge, why is it never used to train moder DNN, like (LLM, ResNet)?

Solution: Complexity high $O(N^3)$.

Question 5: Jacobian vs gradient

What is the fundement diff Betn Jacobian and gradient?

Solution:

- Gradient vector is strictly for scalar valued funtion (many input $\longrightarrow$ one output)
- Jacobian vector is for vector valued function. (many input $\longrightarrow$ many output).

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